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ABC transporter Pdr5 is required for cantharidin resistance in *Saccharomyces cerevisiae*

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ABSTRACT

Cantharidin is a potent anti-cancer drug and is known to exert its cytotoxic effects in several cancer cell lines. Although we have ample knowledge about its mode of action, we still know a little about cantharidin associated drug resistance mechanisms which dictates the efficacy and cytotoxic potential of this drug. In this direction, in the present study we employed *Sacharomyces cerevisiae* as a model organism and screened mutants of pleiotropic drug resistance network of genes for their susceptibility to cantharidin. We show that growth of *pdr1Δ* and *pdr1Δpdr3Δ* was severely reduced in presence of cantharidin whereas that of *pdr3Δ* remain unaffected when compared to wildtype. Loss of one of the *PDR1* target genes *PDR5*, encoding an ABC membrane efflux pump, rendered the cells hypersensitive whereas overexpression of it conferred resistance. Additionally, cantharidin induced the upregulation of both *PDR1* and *PDR5* genes. Interestingly, *pdr1Δpdr5Δ* double deletion mutants were hypersensitive to cantharidin showing a synergistic effect in its cellular detoxification. Furthermore, transcriptional activation of *PDR5* post cantharidin treatment was majorly dependent on the presence of Pdr1 and less significantly of Pdr3 transcription factors. Altogether our findings suggest that Pdr1 acts to increase cantharidin resistance by elevating the level of Pdr5 which serves as a major detoxification safeguard under CAN stress.

1. Introduction

In *Saccharomyces cerevisiae*, pleiotropic drug response (PDR) is a well-documented phenomenon [1,2]. A similar multidrug tolerance phenotype could be observed in human cancer cells and is referred to as multiple drug resistance (MDR) [3,4]. PDR is a general drug response that activates the expression of ATP-binding cassette (ABC) transporter, which can export toxic substances out of the cell thereby allowing the cells to survive [5–9]. The genes involved in PDR are regulated by two main zinc finger transcription factors-*PDR1* and *PDR3* [9,10]. Pdr1p and Pdr3p share ~36% amino acid identity [10]. They can form homo- or heterodimers and act by binding to the PDR elements (PDRE) that are present in the promoter region of their target genes [11]. Targets of Pdr1 and Pdr3 include *PDR5*, *PDR10*, *PDR15*, *YOR1*, *SNQ2* and *TPO1* genes that encode for respective membrane bound ABC transporters [5,12,13]. These transporters facilitate drug efflux from the cells and provide resistance to many xenobiotic compounds including anticancer drugs, fungicides, herbicides, steroids, etc [5,14]. Among these, Pdr5p, a homodimer, is one of the well characterized ABC type protein localized to the plasma membrane and acts as a drug efflux pump [5,15].

Studies indicate that loss of function mutations of either *PDR1* or *PDR3* results in differential drug tolerance whereas loss of both *PDR1* and *PDR3* results in severe drug hypersensitivity [10,16]. In addition, *pdr1Δ* cells have been reported to strikingly reduce yeast resistance to different drugs while the effect of *pdr3Δ* is less severe [17]. Genetic interactions were also shown to exist between *PDR1* and *PDR5* loci. Furthermore, gain-of-function alleles of *PDR1* and *PDR3* are reported to exhibit pleiotropic drug resistance phenotypes [18]. In such drug-resistant *PDR* mutants, expression of the drug efflux pumps Pdr5 is dramatically increased [16], and it has been shown that Pdr1-mediated cycloheximide resistance requires the presence of functional PDR5 [9]. The PDR5 gene disruptant mutant has also been reported to markedly reduce the cellular chloramphenicol [19].

Cantharidin (CAN) is a natural anti-cancer molecule found in blister beetles [20]. It exerts its effects through different

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mechanisms such as inhibition of protein phosphatase 2A, induction of cell cycle arrest, DNA damage, apoptosis, missorting of GPI anchored protein, etc [20,21]. In yeast, *CRG1* is known to methylate CAN and render it inactive [22]. Since the time of its discovery, this drug has been very well exploited to unravel its genetic targets and identify the mechanisms behind its cytotoxicity. But its efficacy, like several other anti-cancer drugs, is influenced by the drug resistance mechanisms such as PDR. So, in the present study we intended to identify the PDR genes responsible for conferring resistance against CAN stress. Our initial growth phenotype assays demonstrate that disruption of *PDR1* and *PDR5* increases CAN toxicity. Also, double gene disruptants of *PDR1*/*PDR3* and *PDR1*/*PDR5* exhibited hypersensitivity upon exposure to CAN. In addition, yeast cells respond to CAN stress by upregulating the expression of both *PDR1* and *PDR5*. Furthermore, qRT-PCR analysis suggest that increase in *PDR5* levels post CAN stress is majorly under the transcriptional control of *PDR1*. Interestingly, overexpression of *PDR5* provided a remarkable resistance for this drug. Altogether our findings suggest that Pdr5 efflux pumps are upregulated by CAN and plays a crucial role in resistance and cellular detoxification for this drug.

2. Materials and methods

2.1. Strains, plasmids, reagents and growth conditions

The *Saccharomyces cerevisiae* strains used in this study are listed in Table S1. All strains were grown in synthetic complete liquid media (SC; 0.17% yeast nitrogen base [YNB], 0.5% ammonium sulfate, 2% glucose and supplemented with required amino acids) at 30 °C. For solid agar media, 2% Bacto-agar was added to SC liquid media. For yeast transformation, the standard lithium acetate procedure was followed and the transformants were propagated in respective selective media as described previously [23].

Gene knockouts and tagging were performed by polymerase chain reaction (PCR) based methods [24,25] using primers listed in Table S2. All single/double gene knockouts were obtained by directly replacing the gene of interest using the pRS405/pRS403/pRS406 vectors. For the selection of positive recombinants, transformants were plated on SC medium devoid of respective auxotrophic markers. Genomic DNA was isolated from the transformants and the knockouts were verified by PCR using ORF-specific primers listed in Table S2 (Fig. S1).

For C-terminal GFP tagging of *PDR5* gene, the coding region of *GFP-HIS3MX6* cassette was amplified from pFA6a vector series by cassette specific primers containing flanking region of the C-terminal of the *PDR5* ORF. The positive colonies were identified by performing Western blot analysis using anti-GFP antibody.

Plasmids used in this study are listed in Table S3. Media components, and all other reagents used in this study were of molecular biology grade and purchased from Himedia, Sigma–Aldrich, New England BioLabs, Invitrogen, BioRad, and Thermo Fisher Scientific.

2.2. Growth phenotype assays

Spot dilution assay was performed by growing the yeast cells overnight. The following day, equal number of yeast cells (optical density; $OD_{600} = 1.0$) were taken and serially diluted tenfold times (10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}). 3 μl of each dilution was spotted onto agarized plates containing DMSO (control) or CAN (Sigma Cantharidin- C7632). All the plates were incubated at 30 °C for 72 h and then scanned using a HP scanner.

For liquid media growth curve analysis, 0.1 OD of exponentially growing yeast cells were treated with either DMSO or CAN in duplicates and seeded onto a 96-well cell culture plate. OD_{600} was measured at a regular interval of 30 min for indicated time period using an automated plate reader. The values obtained were then quantified and plotted using Graph Pad Prism software.

For colony forming unit (CFU) assay, exponentially growing yeast cells were treated with either DMSO or CAN doses and allowed to grow for 1 h. Post treatment, 1 OD of cells from each sample were harvested, washed twice with milliQ and serially diluted tenfold times. 1000 cells were plated onto SC agar plates in duplicates and incubated at 30 °C for 3 days. The viable colonies were counted and represented as percent yeast cell viability as compared to the control plate.

2.3. RNA extraction and qRT-PCR analysis

Total cellular RNAs were isolated from yeast cells by a common heat/freeze phenol method [26]. RNA obtained was reverse transcribed to cDNA by using iScript cDNA Synthesis Kit (BioRad, Catalog 1708891). The cDNA was then used as template for RT-PCR as previously described [27]. The data obtained was then analyzed by $2^{-\Delta\Delta CT}$ method and plotted onto graph using graph pad prism.

2.4. Western blot analysis

Yeast whole cell protein extracts were isolated by using 20% Trichloroacetic acid (TCA) precipitation method [28]. The protein extracts were analyzed by running SDS-PAGE gel (4–20% gradient gel; BioRad) and then transferring the protein to nitrocellulose membrane (NM) following standard electrophoresis methods. Post transfer, the NM was blocked for 1 h using blocking buffer (3% Bovine serum albumin in TBST; Tris buffer saline containing 0.05% Tween-20). The membrane was then incubated with primary antibody for 90min. After washing with 1X TBST thrice, the membrane was again incubated with secondary antibody for 45min. The blots were washed with 1X TBST thrice and left for air drying and scanned using Odyssey infrared imager (LI-COR Biosciences). Primary antibodies used in this study are: anti-GFP (Sigma, Catalog G1544), anti-Tbp (Polyclonal antibodies against recombinant Tbp were raised in rabbit). Goat anti-rabbit (Invitrogen, Catalog A32734) secondary antibody was used.

2.5. Fluorescence microscopy analysis

Exponentially growing PDR5-GFP tagged BY4741 cells were treated with DMSO or CAN 100 μM for 1 h. Post treatment, cells were harvested by centrifugation at 3000 rpm for 3min, washed twice with 1X phosphate buffer saline (PBS; pH-7.4) and imaged using 100x oil-immersion objective lens of ZEISS ApoTome.2 microscope. The subcellular localization of PDR5-GFP fusion protein was monitored using EGFP filter.

2.6. Statistical analysis

All data are representative of 3 independent biological repeats and are presented as mean ± standard error mean (SEM). The significance of differences between sets of data was determined by student t-test; ns denotes $P > 0.05$, * denotes $P \le 0.05$ and ** denotes $P \le 0.01$ (as per the graph pad prism).

3. Results

3.1. Disruption of PDR1 increases CAN mediated toxicity

PDR1 and *PDR3* are two major transcription regulating genes required for pleotropic drug response in yeast [9]. Null mutation of either *PDR1, PDR3* or both of the genes have been reported to confer sensitivity to various drugs or cytotoxic compounds [10,16]. With this premise, we initially performed dosage screening for CAN (Fig. 1A and S2) and generated *pdr1Δ*, *pdr3Δ* and *pdr1Δpdr3Δ* mutants. The WT and mutants were then allowed to grow under CAN stress in both solid and liquid culture media. We found that *pdr1Δ* cells were significantly sensitive to CAN whereas *pdr3Δ* cells remain unaffected when compared to WT cells. Interestingly, *pdr1Δpdr3Δ* mutants were hypersensitive to CAN stress (Fig. 1B). A very similar pattern of toxicity could be seen in liquid media growth curve analysis also (Fig. 1C). Additionally, a fivefold increase in the *PDR1* gene expression was observed post exposure of wild type BY4741 cells to CAN 100 μM for 1 h (Fig. 1D). So, the comparison of susceptibility of wild type cells and mutants to CAN revealed that *PDR1*, but not *PDR3,* is crucial for CAN tolerance.

3.2. ABC transporter Pdr5 is a key determinant of yeast adaptation to CAN

PDR1 has been shown to regulate the expression of several downstream genes encoding for multidrug transporters like Pdr5, Yor1, Snq2, etc [13]. Disruption of one or many such transporters simultaneously have been reported to confer sensitivity to various drugs [5,14,17]. Thus, we next investigated the effect of CAN on Pdr1 target genes. To achieve this, single gene disruptant mutant of six genes (*PDR5, PDR10, PDR15, YOR1, SNQ2* and *TPO1*) known to be regulated by *PDR1* were generated and their susceptibility to CAN was compared with the WT strain. It was observed that only the growth of mutant deleted for gene *PDR5* (*pdr5Δ*) was severely reduced on plates containing CAN whereas the growth of remaining mutants remained unaffected (Fig. 2A). These strains were also assayed for their growth phenotype in liquid media wherein only *pdr5Δ* strain was found to be highly sensitive (Fig. S3). These data suggest that absence of *PDR5* leads to an enhanced CAN toxicity.

As deletion of *PDR5* significantly reduced the tolerance level for CAN, we next wanted to check how the *PDR5* expression alters in response to CAN stress. To achieve this, we first performed qRT-PCR analysis and measured the relative mRNA expression levels of *PDR5* post CAN 50, 100 and 150 μM treatment for 1 h. There was a significant increase in the *PDR5* gene expression when cells were exposed to CAN (100 and 150 μM) as compared to the control cells (Fig. 2B). With increasing concentration of CAN and cytotoxicity we could observe a concomitant upregulation in *PDR5* gene expression. We next employed the fluorescence microscopy technique to visualize the Pdr5 protein levels (*PDR5* C-terminally tagged with GFP) inside the cells. As shown in Fig. 2C, cells treated with CAN 100 μM for 1 h have more Pdr5 protein levels than the untreated ones. To further substantiate this finding, the cells harvested from similar set up as above were subjected to Western blot analysis and a very significant increase in the Pdr5-GFP fusion protein level post CAN 100 μM treatment was obtained (Fig. 2D). Overall, these findings suggest that Pdr5 is a key determinant of yeast adaptation to CAN.

3.3. Overexpression of PDR5 alleviates CAN toxicity

As absence of functional *PDR5* leads to increased CAN toxicity we next tested whether overexpression of *PDR5* can help cells tolerate toxic levels of CAN. In this direction, we transformed the wild type and *pdr5Δ* cells with a 2-μm overexpression PDR5

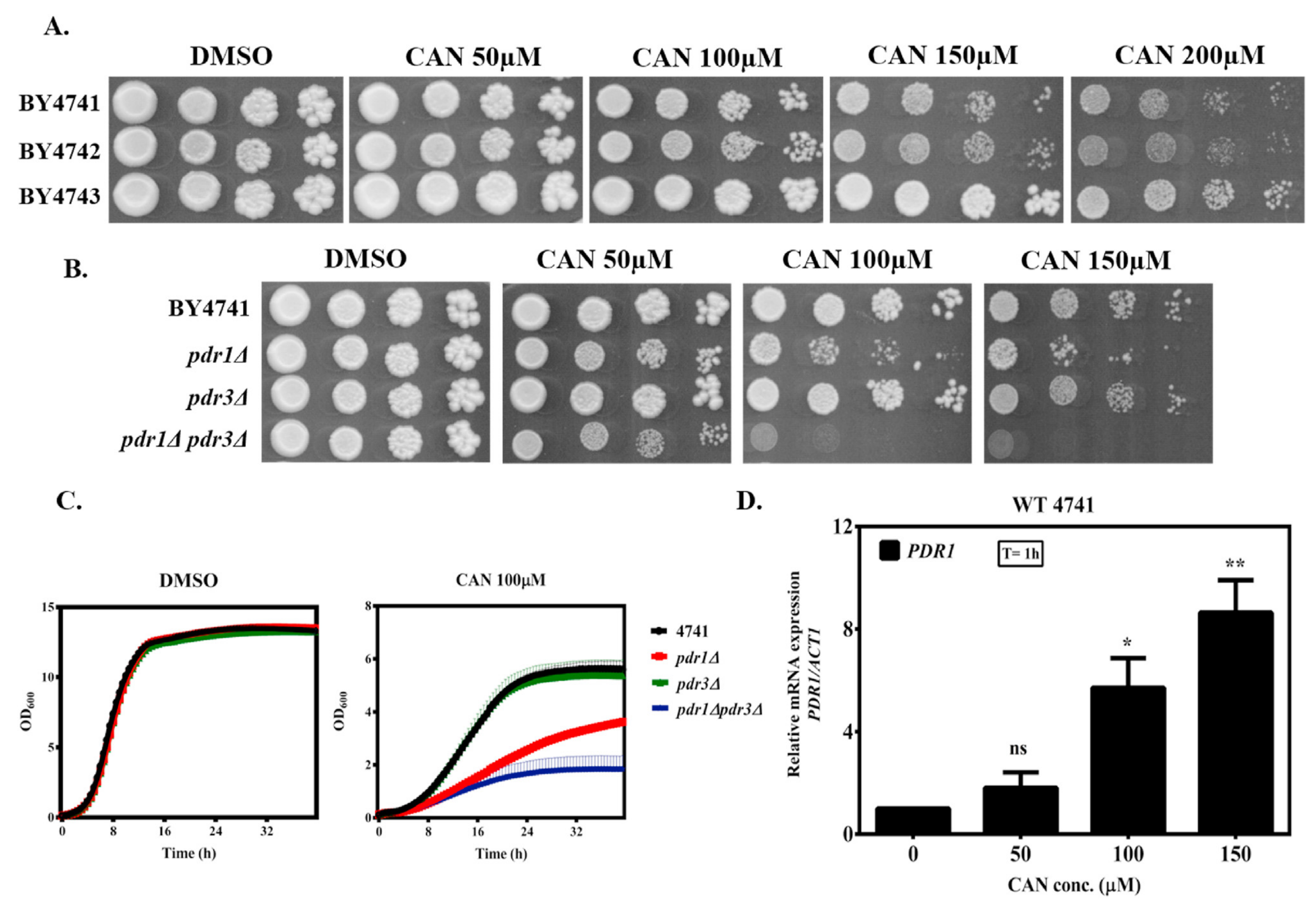

Fig. 1. ***PDR1* is required for CAN stress tolerance. (A)** Yeast cells are sensitive to CAN in a dose-dependent manner **(B)** *pdr1Δ* and *pdr1Δpdr3Δ* cells are sensitive to CAN stress **(C)** Growth curve analysis in liquid media was done using indicated strains for 36 h **(D)** CAN upregulated the expression of *PDR1.*

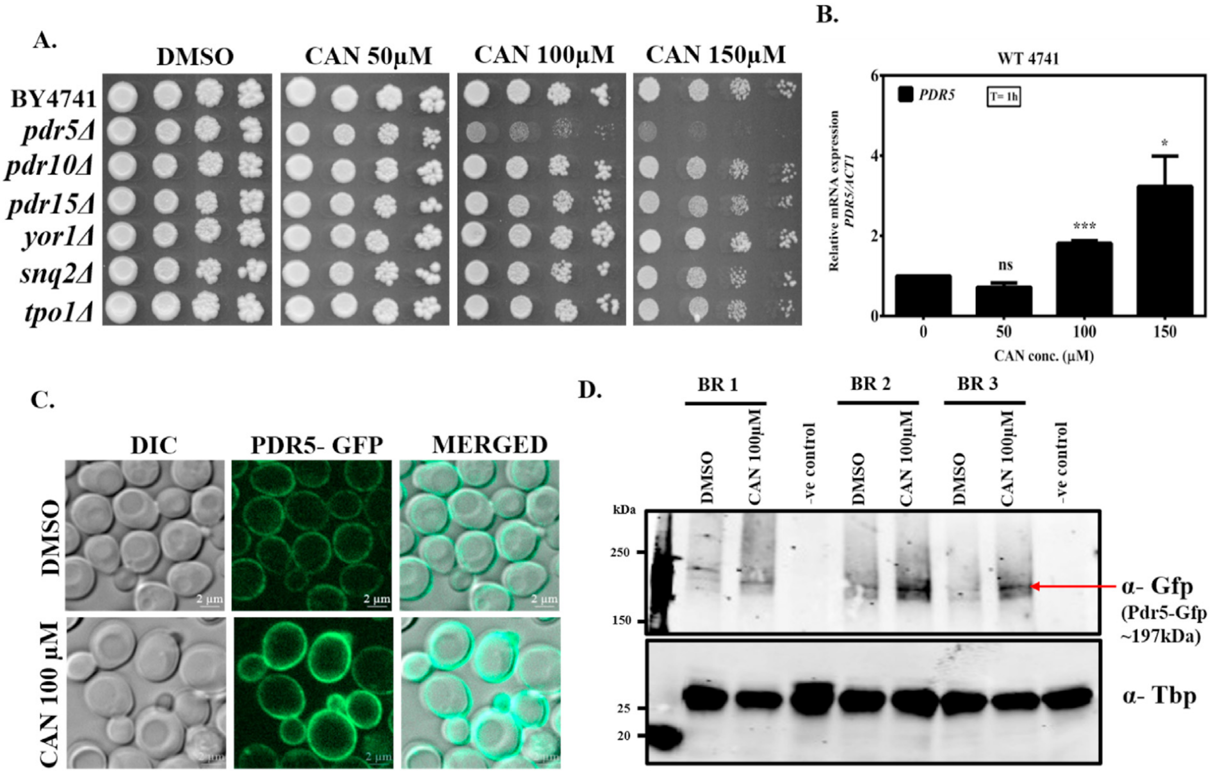

Fig. 2. CAN induces the expression of *PDR5* (A) *pdr5Δ* cells are hypersensitive to CAN **(B)** CAN upregulated the expression of *PDR5* **(C)** Fluorescence microscopy analysis to show that CAN induces the PDR5-GFP levels. Scale bar-2μm **(D)** Western blot analysis of Pdr5 protein using anti-GFP antibody.

plasmid and its corresponding empty vector (pYEp13). Serial dilutions of independent transformants were assayed for growth in the presence or absence of CAN on selective plates lacking leucine. As shown in Fig. 3, cells harboring the *PDR5* overexpression plasmid were able to grow significantly better than the ones harboring empty vector across all CAN containing plates. In addition, overexpression of *PDR5* provided a remarkable resistance against very toxic levels of CAN 300 μM. In liquid media also overexpression of *PDR5* relieved the toxic effect of CAN (Fig. S4). Taken together, these results suggest that yeast cells mitigate CAN toxicity by elevating the levels of Pdr5 efflux pump.

3.4. Cantharidin mediated upregulation of PDR5 is majorly under the control of PDR1

So far, we have shown that *PDR1* and *PDR5* genes are required for CAN stress tolerance. From literature it is known that *PDR5* is under the transcriptional control of *PDR1* and *PDR3* [10]. So, we created a double deletion mutant (*pdr1Δpdr5Δ*) to check whether these two genes act synergistically to modulate CAN toxicity or not.

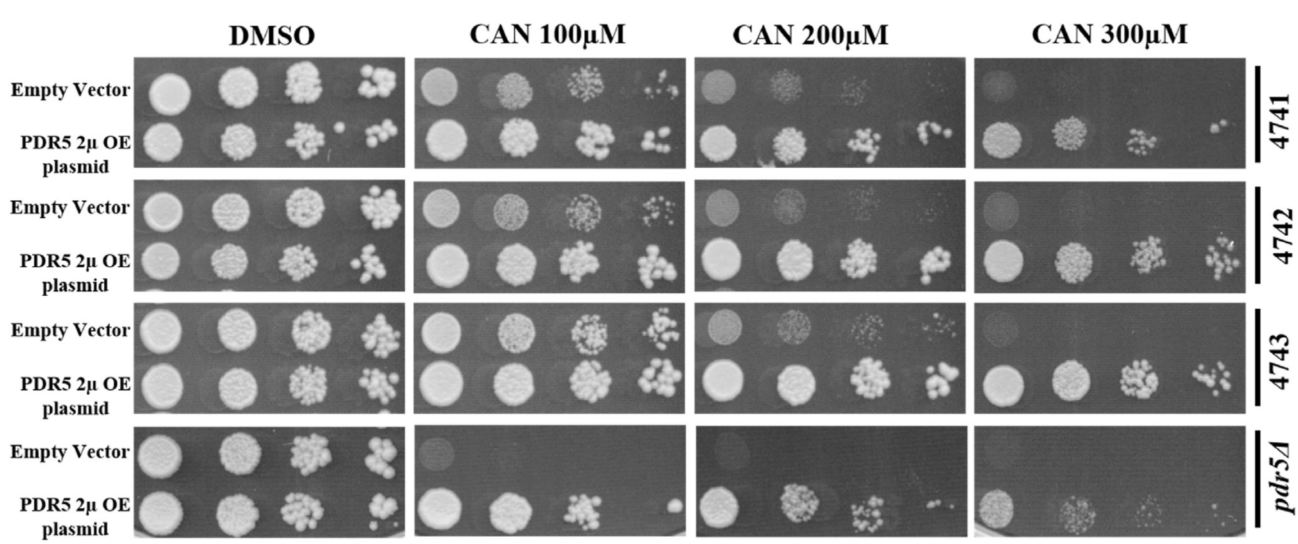

Fig. 3. Overexpression of *PDR5* confers CAN resistance. Wild type and *pdr5Δ* cells harboring a *PDR5* 2μ overexpressing plasmid could tolerate very toxic levels of CAN.

Spot assay data revealed that *pdr1Δpdr5Δ* was hypersensitive to CAN than *pdr1Δ* and *pdr5Δ* mutants (Fig. 4A). We also observed *pdr3Δpdr5Δ* and *pdr5Δ* mutants to be equally sensitive in presence of CAN. A very similar pattern of growth could be seen in liquid media as well (Fig. S5). These observations suggest that combination of two gene deletions i.e., *PDR1* and *PDR5 (pdr1Δpdr5Δ),* but not *PDR3* and *PDR5*, significantly compromise yeast cell viability in presence of CAN as compared to the single gene deletions (*pdr1Δ* or *pdr3Δ* or *pdr5Δ*). This suggests the synergistic role of both Pdr1 and Pdr5 in CAN tolerance.

We also examined whether the increase in *PDR5* gene expression post CAN treatment is *PDR1* or *PDR3* dependent. To ascertain this, we performed qRT-PCR analysis to measure the transcript levels of *PDR5* in BY4741, *pdr1Δ, pdr3Δ* and *pdr1Δpdr3Δ* cells under CAN 100 μM stress. As shown in Fig. 4B, the activation of *PDR5* transcription post CAN treatment is significantly abolished in the *pdr1Δ* cells and it is comparable to wildtype in *pdr3Δ* cells. Whereas absence of both *PDR1* and *PDR3 (pdr1Δpdr3Δ)* resulted in a further decrease in *PDR5* activation compared to that of *pdr1Δ* cells. The lower level of activation of *PDR5* gene transcription in *pdr1Δ* cells compared to *pdr3Δ cells* is consistent with the more subtle effect that *PDR3* deletion exerts on yeast susceptibility to CAN. This suggests that *PDR5* transcriptional activation in presence of CAN is majorly under the control of *PDR1.*

4. Discussion

Cantharidin is a well exploited anti-cancer molecule [20]. Although we have ample knowledge about mode of action and genetic targets of CAN, we are unaware of the multidrug resistance mechanisms associated with this drug. Understanding MDR mechanisms is of utmost importance in cancer therapeutics as extrusion of drugs via membrane transporters reduces drug efficacy and increases drug resistance [3,4,29–31]. In cancer cells, overexpression of ABC transporters such as P-glycoprotein and multidrug resistance-associated protein have been implicated in conferring resistance to chemotherapeutic drugs [30,32]. In this direction, budding yeast *Saccharomyces cerevisiae* has served as an invaluable model organism for the genetic identification of multidrug resistance mechanisms [2]. In yeast, increased expression of ABC type efflux-pumps results in expulsion of drug out of the cell [7,33]. This in turn leads to reduced intracellular accumulation and potency of drugs [2,34]. Till date, ABC transporters such as Pdr5 (ortholog of mammalian MDR1), Yor1 and Snq2 have been reported to protect yeast cells from a broad range of drugs [7,14]. These transporters confer increased resistance when overexpressed probably by facilitating a higher amount of drug efflux [14]. Other determinants of drug resistance include transcription regulators such as *PDR1* and *PDR3*, which control the activation of such ABC transporters [2].

With this premise, in the present study, we intended to analyze the molecular mechanisms of yeast susceptibility to CAN using different mutants of the PDR network of genes involved in yeast multidrug resistance. To achieve this, we first performed a dosage screening and found that yeast cells were sensitive to CAN in a dose dependent manner. Growth phenotype assays suggested that loss of *PDR1* (a master transcription regulator of ABC transporters) but not *PDR3* increased yeast susceptibility to CAN whereas loss of both *PDR1* and *PDR3* proved to be lethal. This indicated that *PDR1* (major contribution) and *PDR3* (minor contribution) act in concert to modulate CAN toxicity. As *pdr1Δ* was highly sensitive, we next examined the effect of CAN on Pdr1p regulated genes and found that loss of one of its target genes *PDR5* (an important ABC transporter) rendered the cells sensitive. In order to study expression

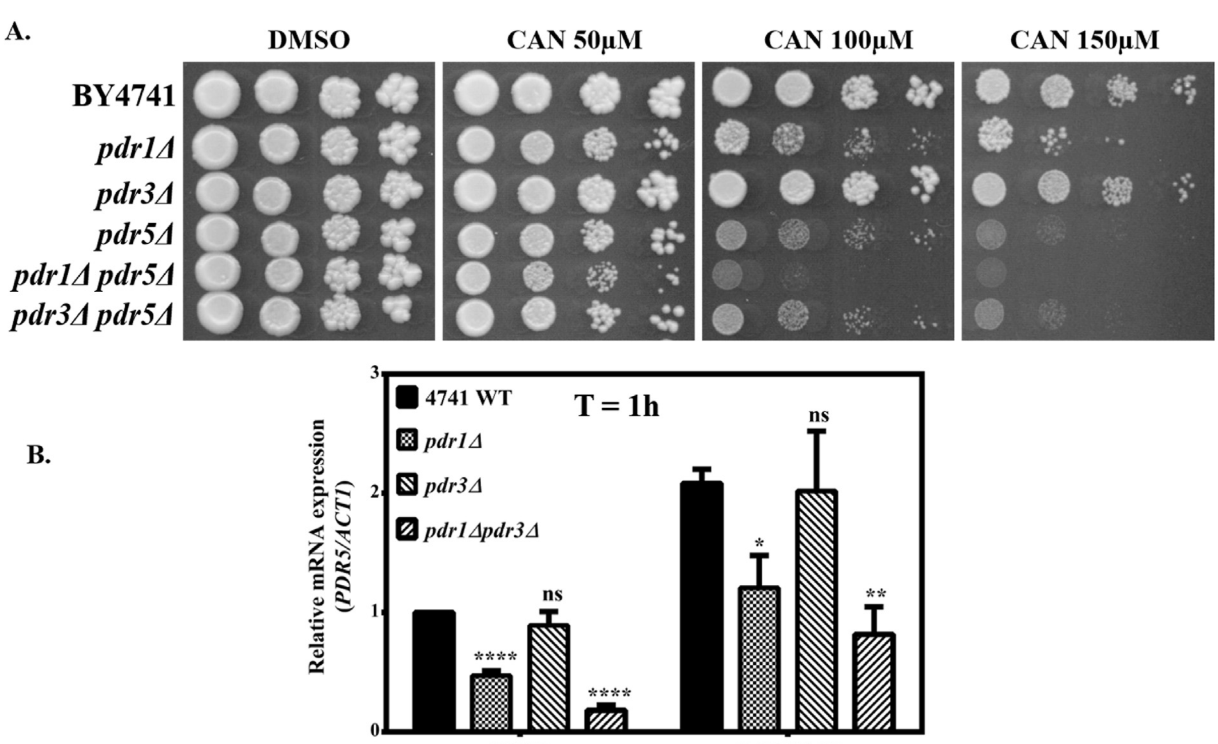

Fig. 4. Role of *PDR1* in transcriptional activation of *PDR5* under CAN stress. (A) *pdr1Δpdr5Δ* cells exhibit hypersensitivity **(B)** Comparison of relative mRNA levels of *PDR5* in BY4741, *pdr1Δ, pdr3Δ* and *pdr1Δpdr3Δ* cells.

patterns of the *PDR1* and *PDR5* genes, we incubated cells with a sublethal dose of CAN for 1 h and determined the transcript levels of each gene by qRT-PCR approach. Results suggest that, there was a significant induction in the gene expression level of both the genes (*PDR1*- 5-fold; *PDR5*- 3-fold) post CAN treatment. To further validate this finding, microscopic and Western blot analysis was done to check the protein levels of Pdr5 (Pdr5-GFP). Data showed a significant increase in the protein levels of Pdr5 under CAN stress. Furthermore, overexpression of *PDR5* helped the yeast cells tolerate high toxic levels of CAN. With increasing concentration of CAN its associated stress/toxicity increases which inturn leads to upregulation of Pdr5 required for cellular detoxification. These results indicate that CAN-inducible Pdr5p is the primary membrane transporter responsible for CAN resistance.

Additionally, deletion of *PDR1* in a *pdr5Δ* background dramatically potentiates the sensitivity observed in a *pdr5Δ* strain. On the other hand, deletion of *PDR3* in a *pdr5Δ* background showed similar susceptibility to that observed in a *pdr5Δ* strain. These findings suggested that Pdr1 (not Pdr3) and Pdr5 act synergistically to confer CAN resistance.

Next, we investigated the role of *PDR1* and *PDR3* in transcriptional activation of *PDR5* under CAN stress. To achieve this, we performed qRT-PCR analysis and compared the gene expression level of *PDR5* across BY4741, *pdr1Δ, pdr1Δ* and *pdr1Δpdr3Δ* strains. We found that *PDR5* gene expression decreases significantly in *pdr1Δ* strain, whereas in *pdr3Δ* it does not change the expression when compared to wild type levels. However, *PDR5* gene transcription was highly reduced in *pdr1Δpdr3Δ* double deletion mutant as compared to both WT and *pdr1Δ* levels. Interestingly, this pattern of *PDR5* gene expression remained the same across WT and mutants either in the presence or in the absence of CAN. In agreement with the previous data obtained from growth phenotype assays, these data show that in comparison to *PDR3*, *PDR1* contribution to transcriptional activation of *PDR5* seems to be maximal.

In summary, we identified *PDR5* as a main CAN resistance gene. We also observed a synergistic effect of Pdr1 and Pdr5 in cellular detoxification of CAN. And activation of Pdr5-mediated resistance of CAN is majorly under the transcriptional control of Pdr1. As mammalian MDR is analogous to yeast PDR [7], the resistance mechanisms reported with CAN here can be extrapolated to cancer cells.

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Declaration of competing interest

The authors declare no competing financial interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2021.03.074.

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